A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and complementary causal analysis

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ-VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the convolutional 20.1M-parameter instantiation reaches $93.7\% \pm 0.8\%$ under the published three-seed protocol. Averaging its three fixed checkpoints reaches 96.2% at three times the inference cost. In a separate three-checkpoint ViT instantiation, we derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module with worst relative error 2.56×10^{-7}. All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. In an exploratory rank screen at the ViT visual bond, retaining 96 of 384 causally ranked dimensions succeeds in $50/60$ matched trials across three checkpoints, compared with $52/60$ for the full representation. Three equal-rank random controls reach only $1/60$, $3/60$, and $3/60$. The selected subspace retains 49 of the 52 full-policy successes. On disjoint cache samples, it has 33.5–$110.6\times$ lower reconstruction error than the median across three random projectors, computed separately by action group. These are complementary analyses of one rational model family. They do not yet map the exact attention factors directly to the causal visual basis. The evidence establishes strong specialist capability, exact layerwise access, and a compact exploratory policy bottleneck. It does not establish broad robotic generalization or robust instruction grounding.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

> *Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis and causal representation tests?*

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit and causal representation screen.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. In a separate exploratory rank screen, a selected 96-dimensional visual subspace retains 49 of 52 full-policy successes across three checkpoints. Three equal-rank random controls reach only $1/60$, $3/60$, and $3/60$.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible and supports separate causal tests of its internal representations.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho}L_{\rho i}R_{\rho j}\right)\bar{x}_i\bar{x}_j. \tag{1}$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^{\top}) \odot (Q_2 K_2^{\top})/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network.

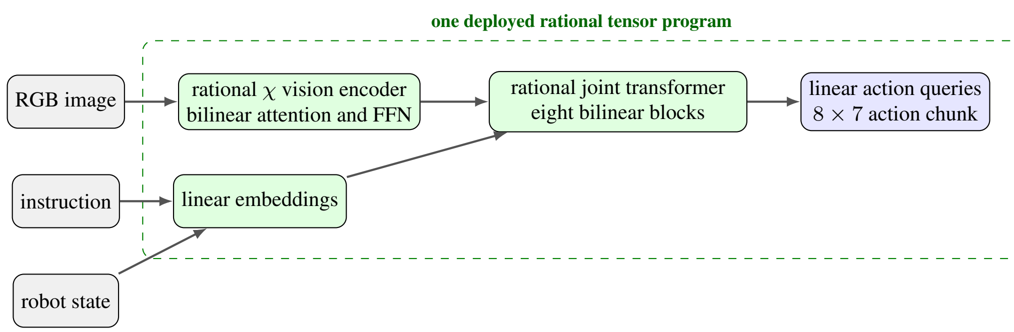

Figure 1: **Deployed χ-VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

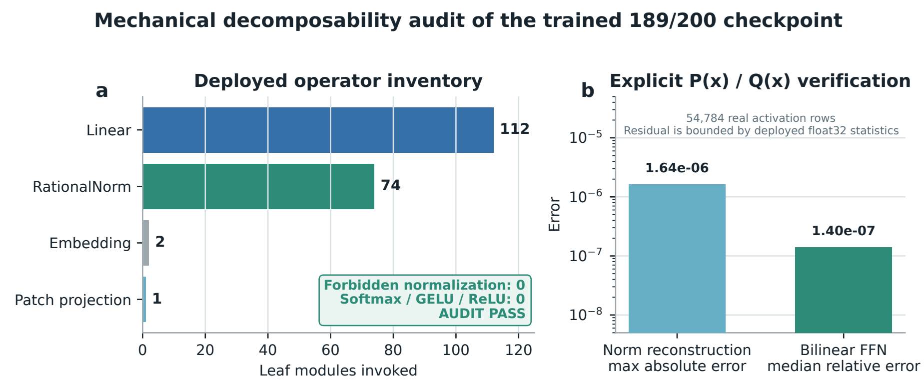

Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and $1{,}500$ rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

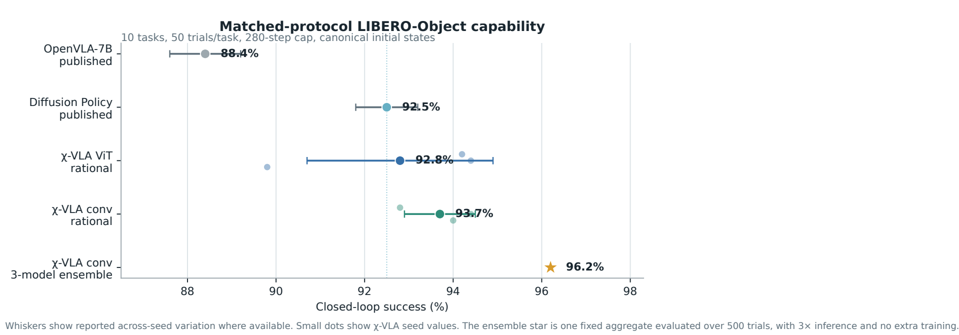

Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7}\% \pm \mathbf{0.8}\%$
χ-VLA, conv ensemble	3 × 20.1M	$3 \to 1$	$\mathbf{96.2}\%$
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14}.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer's computation.

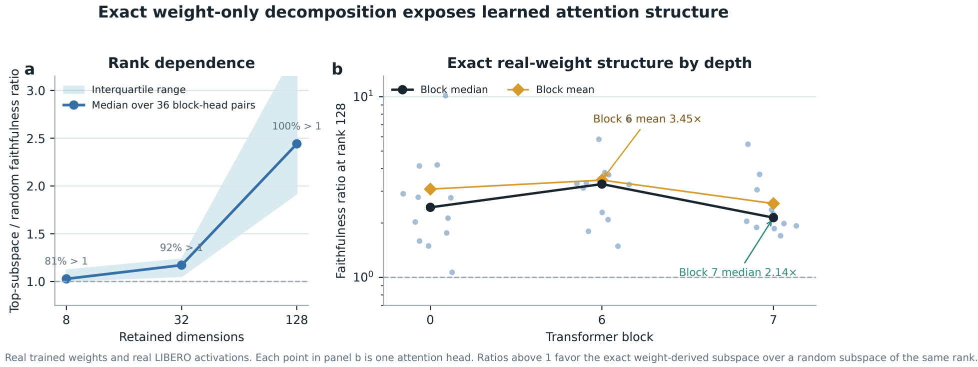

Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure

We next test the policy's representation with a separate causal analysis.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we build one shared action-balanced ranking. We form one Jacobian Gram per action coordinate, trace-normalize the seven Grams, and average them. A subspace is a smaller coordinate system inside the model's 384-dimensional visual representation. Retaining a subspace means zeroing every direction outside it, then running the actual policy. We compare the selected directions with random equal-width controls in offline reconstruction and closed-loop intervention. This ranking uses cached training demonstrations and downstream action gradients. Offline reconstruction uses a disjoint set of cache samples that did not estimate the ranking. Both splits still come from the training cache, so this is a disjoint-sample in-cache diagnostic. The closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result.

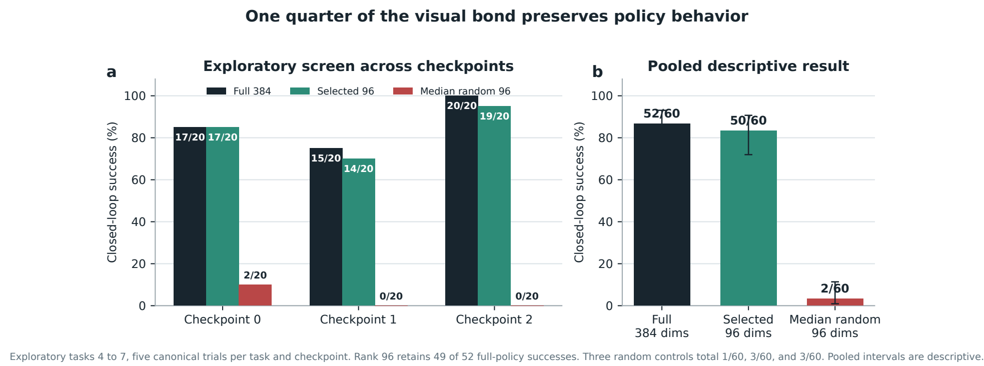

Figure 5: **One quarter of the visual bond preserves policy behavior in the exploratory rank screen.** Each checkpoint uses four tasks and five paired canonical trials per task. Rank 96 retains 49 of 52 full-policy successes. Three equal-rank random controls reach $1/60$, $3/60$, and $3/60$. The pooled interval is descriptive because trials within a checkpoint share learned weights.

The exploratory action-balanced screen tests ranks 64, 96, 128, and 192 on tasks 4 to 7. Appendix Figure 6 reports the full curve. Rank 96 is the smallest tested width with more than 90% conditional retention while random success remains below 5%. Tasks 8 and 9 are reserved for a frozen confirmation and are not used here. At rank 96, full representations succeed in $17/20$, $15/20$, and $20/20$ trials across checkpoints. Selected subspaces reach $17/20$, $14/20$, and $19/20$, and retain 49 of the 52 exact full-policy successes. Their conditional retention is 94.1%, 93.3%, and 95.0%. Three random controls per checkpoint total only $1/60$, $3/60$, and $3/60$. Every checkpoint and random-control comparison has the same selected-over-random direction. Within each fixed checkpoint, the descriptive exact paired tests have $p < 2.5 \times 10^{-4}$. Offline fidelity uses 1,024 discovery samples and 1,024 disjoint evaluation samples per checkpoint. Against the median across three random projectors, computed separately by action group, the selected basis has 53.0–110.6$\times$ lower translation error, 37.5–62.1$\times$ lower rotation error, and 33.5–103.0$\times$ lower gripper error. This repairs sample reuse, but it remains an in-cache fidelity test rather than held-out task generalization. This data-driven full-policy ranking complements the exact layerwise analysis in Section 4.

6 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ-VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ-nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact exact weight auditing and an exploratory causal bottleneck as complementary analyses in a strong continuous vision-language policy.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist capability setting rather than evidence of broad generalization.

7 Limitations and conclusion

χ-VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers, and a compact causal visual subspace retains 49 of 52 full-policy successes across three checkpoints while equal-rank random controls rarely succeed.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions, so these results do not establish broad robotic generalization or robust instruction grounding. Exact reconstruction is certified layer by layer, while the full-policy visual ranking also uses cached activations and action gradients. The exact attention factors and causal visual basis are complementary analyses, not a demonstrated direct mapping. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

Within that scope, polynomial and rational restrictions are compatible with strong closed-loop control and provide a precise weight-derived analysis surface. Convolutional checkpoints support the capability result, while separate ViT checkpoints support the exact reconstruction and causal-bottleneck analyses. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Rank screen and selectivity

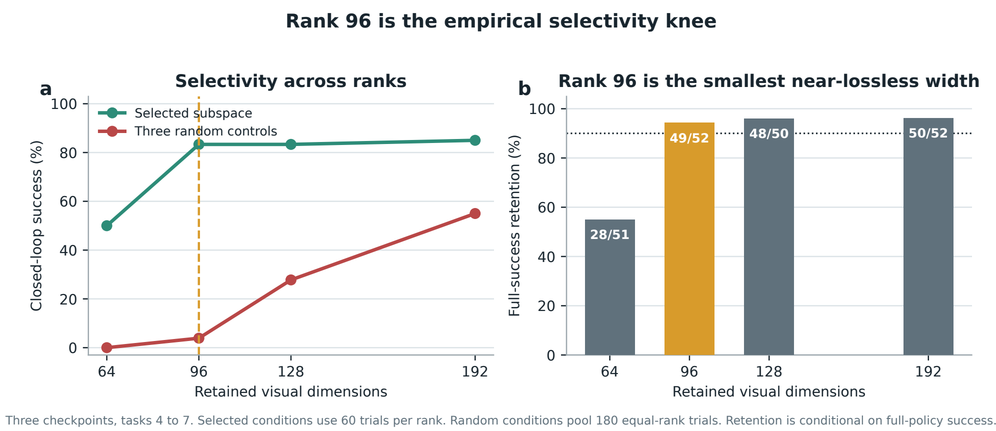

Figure 6: **Rank** 96 **is the empirical selectivity knee.** Selected subspaces use 60 closed-loop trials per rank across three checkpoints. Random results pool three equal-rank controls per checkpoint, for 180 trials per rank. Each rank has its own paired full-policy condition. Rank 96 is the smallest width above 90% conditional retention, while random success remains below 5%.

The selected subspace rises from $30/60$ successes at rank 64 to $50/60$ at rank 96. Wider ranks add little selected success, reaching $50/60$ and $51/60$ at ranks 128 and 192. In contrast, random success rises from $7/180$ at rank 96 to $50/180$ and $99/180$. Conditional retention is 54.9%, 94.2%, 96.0%, and 96.2% across the four ranks. Thus rank 96 preserves nearly all baseline successes before equal-width random subspaces become broadly capable.

A.2 Per-task ensemble behavior

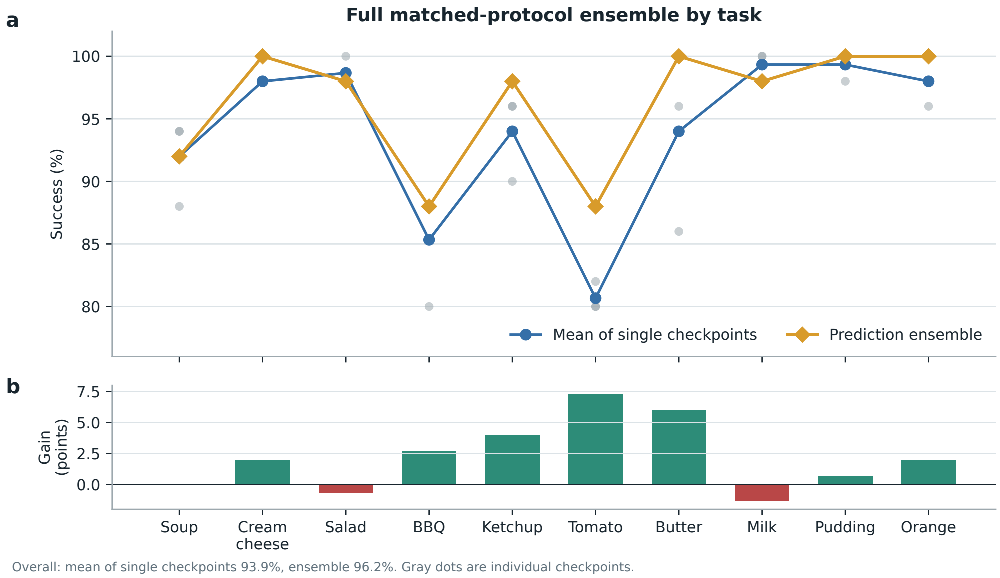

Figure 7: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.3 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.4 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.